4-GPU NVLink starter | 3,200 labelled contacts | 384 connected conductors

Open a GPU sheet, then connector A or B. Identical global labels join the NVLink data pins between GPUs.
Clock, power, ground, PCIe and control contacts remain labelled and unconnected for your own circuitry.

GPU0 / J1 + J2

GPU0 / V100 SXM2
A: J1 / 400 labelled contacts
B: J2 / 400 labelled contacts
192 NVLink data contacts wired

File: GPU0.kicad_sch

GPU1 / J3 + J4

GPU1 / V100 SXM2
A: J3 / 400 labelled contacts
B: J4 / 400 labelled contacts
192 NVLink data contacts wired

File: GPU1.kicad_sch

GPU2 / J5 + J6

GPU2 / V100 SXM2
A: J5 / 400 labelled contacts
B: J6 / 400 labelled contacts
192 NVLink data contacts wired

File: GPU2.kicad_sch

GPU3 / J7 + J8

GPU3 / V100 SXM2
A: J7 / 400 labelled contacts
B: J8 / 400 labelled contacts
192 NVLink data contacts wired

File: GPU3.kicad_sch

BIDIRECTIONAL GROUP PAIRINGS

GPU0 / NVHS0 <-> GPU1 / NVHS3

GPU0 / NVHS1 <-> GPU1 / NVHS2

GPU2 / NVHS0 <-> GPU3 / NVHS3

GPU2 / NVHS1 <-> GPU3 / NVHS2

GPU0 / NVHS4 <-> GPU2 / NVHS5

GPU0 / NVHS5 <-> GPU2 / NVHS4

GPU1 / NVHS4 <-> GPU3 / NVHS5

GPU1 / NVHS5 <-> GPU3 / NVHS4

GPU0 / NVHS2 <-> GPU3 / NVHS1

GPU0 / NVHS3 <-> GPU3 / NVHS0

GPU1 / NVHS0 <-> GPU2 / NVHS3

GPU1 / NVHS1 <-> GPU2 / NVHS2

DATA WIRING ONLY — NOT A COMPLETE WORKING NVLINK IMPLEMENTATION

External NVLink clock required on EVERY V100: 156.25 MHz differential / Connector B E20=P, F20=N.
Separate PCIe clock: 100 MHz differential / Connector A E7=P, F7=N. Clock circuits are not included.
Use suitable source/fanout outputs per GPU; also provide the required supplies, reset and supporting controls.

REFERENCE SCOPE

Four-GPU logical plan allocates two NVHS groups per peer.
This complete topology is not claimed as hardware-tested. Physical connector orientation and mating footprints require confirmation.
Curated from the credited community sources; see the repository sources, connection guide and standalone workbook.

Logical NVLink reference; no hardware qualification or footprint assignment
Pigeon Labs

Sheet: /		
File: V100_4GPU_NVLink.kicad_sch		
Title: 4 V100 SXM2 — NVLink reference starter		
Size: A3	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 1/13

GPU0: 800 labelled contacts / 192 NVLink data contacts connected
Open Connector A for PCIe, clock and power/control contacts; open B for the NVLink net labels.

GPU0 Connector A / J1



File: GPU0_Connector_A.kicad_sch

GPU0 Connector B / J2



File: GPU0_Connector_B.kicad_sch

Supporting clock, supply and control pins remain labelled and unconnected.
Contact symbols are passive and have no assigned PCB footprint.

CLOCK CIRCUITRY REQUIRED FOR OPERATION
NVLink: 156.25 MHz differential / B:E20=P, B:F20=N
PCIe: 100 MHz differential / A:E7=P, A:F7=N

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU0 / J1 + J2/
File: GPU0.kicad_sch

Title: GPU0 — V100 SXM2 connectors

Size: A4	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 2/13

All 400 contacts retain their signal names and pad numbers. Clocks, supplies, PCIe, reset and control circuits are left for your design.

J1A

A1	GND
A2	PERp0
A3	PERn0
A4	GND
A5	PERp7
A6	PERn7
A7	GND
A8	PERp8
A9	PERn8
A10	GND
A11	PERp12
A12	PERn12
A13	GND
A14	GND
A15	UNRESOLVED/NC
A16	UNRESOLVED/NC
A17	GND
A18	PWR_GOOD
A19	GND
A20	+5V_AUX
A21	GND
A22	12V
A23	12V
A24	GND
A25	12V
A26	12V
A27	GND
A28	12V
A29	12V
A30	GND
A31	12V
A32	12V
A33	GND
A34	12V
A35	12V
A36	GND
A37	12V
A38	12V
A39	GND
A40	12V

GPU0 / A / Column A

J1B

B1	PERp1
B2	PERn1
B3	GND
B4	PERp4
B5	PERn4
B6	GND
B7	GND
B8	GND
B9	PERp9
B10	PERn9
B11	GND
B12	PERp15
B13	PERn15
B14	GND
B15	GND
B16	UNRESOLVED/NC
B17	UNRESOLVED/NC
B18	GND
B19	GND
B20	+5V_AUX
B21	GND
B22	12V
B23	12V
B24	GND
B25	12V
B26	12V
B27	GND
B28	12V
B29	12V
B30	GND
B31	12V
B32	12V
B33	GND
B34	12V
B35	12V
B36	GND
B37	12V
B38	12V
B39	GND
B40	12V

GPU0 / A / Column B

J1C

C1	GND
C2	PERp3
C3	PERn3
C4	GND
C5	PERp6
C6	PERn6
C7	GND
C8	PERp10
C9	PERn10
C10	GND
C11	PERp13
C12	PERn13
C13	GND
C14	GND
C15	UNRESOLVED/NC
C16	UNRESOLVED/NC
C17	GND
C18	I2C_ADDR0
C19	GND
C20	+5V_AUX
C21	GND
C22	12V
C23	12V
C24	GND
C25	12V
C26	12V
C27	GND
C28	12V
C29	12V
C30	GND
C31	12V
C32	12V
C33	GND
C34	12V
C35	12V
C36	GND
C37	12V
C38	12V
C39	GND
C40	12V

GPU0 / A / Column C

J1D

D1	PERp2
D2	PERn2
D3	GND
D4	PERp5
D5	PERn5
D6	GND
D7	GND
D8	GND
D9	PERp11
D10	PERn11
D11	GND
D12	PERp14
D13	PERn14
D14	GND
D15	GND
D16	UNRESOLVED/NC
D17	UNRESOLVED/NC
D18	GND
D19	GND
D20	+5V_AUX
D21	GND
D22	12V
D23	12V
D24	GND
D25	12V
D26	12V
D27	GND
D28	12V
D29	12V
D30	GND
D31	12V
D32	12V
D33	GND
D34	12V
D35	12V
D36	GND
D37	12V
D38	12V
D39	GND
D40	12V

GPU0 / A / Column D

J1E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	PCIE_REFCLK_P
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	PERST#
E19	GND
E20	+5V_AUX
E21	GND
E22	12V
E23	12V
E24	GND
E25	12V
E26	12V
E27	GND
E28	12V
E29	12V
E30	GND
E31	12V
E32	12V
E33	GND
E34	12V
E35	12V
E36	GND
E37	12V
E38	12V
E39	GND
E40	12V

GPU0 / A / Column E

J1F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	PCIE_REFCLK_N
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	PWR_BRAKE#
F19	GND
F20	JTAG_TDI
F21	GND
F22	12V
F23	12V
F24	GND
F25	12V
F26	12V
F27	GND
F28	12V
F29	12V
F30	GND
F31	12V
F32	12V
F33	GND
F34	12V
F35	12V
F36	GND
F37	12V
F38	12V
F39	GND
F40	12V

GPU0 / A / Column F

J1G

G1	PETp0
G2	PETn0
G3	GND
G4	PETp4
G5	PETn4
G6	GND
G7	GND
G8	GND
G9	PETp8
G10	PETn8
G11	GND
G12	PETp15
G13	PETn15
G14	GND
G15	GND
G16	UNRESOLVED/NC
G17	UNRESOLVED/NC
G18	GND
G19	GND
G20	JTAG_TDO
G21	GND
G22	12V
G23	12V
G24	GND
G25	12V
G26	12V
G27	GND
G28	12V
G29	12V
G30	GND
G31	12V
G32	12V
G33	GND
G34	12V
G35	12V
G36	GND
G37	12V
G38	12V
G39	GND
G40	12V

GPU0 / A / Column G

J1H

H1	GND
H2	PETp1
H3	PETn1
H4	GND
H5	PETp7
H6	PETn7
H7	GND
H8	PETp9
H9	PETn9
H10	GND
H11	PETp12
H12	PETn12
H13	GND
H14	GND
H15	UNRESOLVED/NC
H16	UNRESOLVED/NC
H17	GND
H18	I2C_ADDR1
H19	JTAG_SEL
H20	JTAG_TCLK
H21	GND
H22	12V
H23	12V
H24	GND
H25	12V
H26	12V
H27	GND
H28	12V
H29	12V
H30	GND
H31	12V
H32	12V
H33	GND
H34	12V
H35	12V
H36	GND
H37	12V
H38	12V
H39	GND
H40	12V

GPU0 / A / Column H

J1I

J1	PETp2
J2	PETn2
J3	GND
J4	PETp5
J5	PETn5
J6	GND
J7	GND
J8	GND
J9	PETp11
J10	PETn11
J11	GND
J12	PETp14
J13	PETn14
J14	GND
J15	GND
J16	UNRESOLVED/NC
J17	UNRESOLVED/NC
J18	GND
J19	GND
J20	JTAG_TMS
J21	GND
J22	12V
J23	12V
J24	GND
J25	12V
J26	12V
J27	GND
J28	12V
J29	12V
J30	GND
J31	12V
J32	12V
J33	GND
J34	12V
J35	12V
J36	GND
J37	12V
J38	12V
J39	GND
J40	12V

GPU0 / A / Column J

J1J

K1	GND
K2	PETp3
K3	PETn3
K4	GND
K5	PETp6
K6	PETn6
K7	GND
K8	PETp10
K9	PETn10
K10	GND
K11	PETp13
K12	PETn13
K13	GND
K14	GND
K15	UNRESOLVED/NC
K16	UNRESOLVED/NC
K17	GND
K18	PWR_EN
K19	THERM_OVERT#
K20	JTAG_TRST#
K21	GND
K22	12V
K23	12V
K24	GND
K25	12V
K26	12V
K27	GND
K28	12V
K29	12V
K30	GND
K31	12V
K32	12V
K33	GND
K34	12V
K35	12V
K36	GND
K37	12V
K38	12V
K39	GND
K40	12V

GPU0 / A / Column K

Separate PCIe reference clock: 100 MHz differential on A/E7 (P) and A/F7 (N). Clock circuitry is not included.

A/B7 and A/A39: disputed GND labels (reference presence loop, PRSNT2_A/B).
Ground both only if your carrier omits presence detection and its controller does not depend on it.
For presence detection, implement and validate the loop. Grounding its readout can force a false present state.
Source: Quanta Big Basin sheets 15–16. See docs/presence-detection.md.

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU0 / J1 + J2/GPU0 Connector A / J1/
File: GPU0_Connector_A.kicad_sch

Title: GPU0 — Connector A — 400 contacts

Size: A3	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 3/13

NVLink data pins connect through matching global labels. Clock, ground, power and control contacts keep their names and remain unconnected.

DATA WIRING ONLY — EXTERNAL CLOCK REQUIRED
Each V100: 156.25 MHz differential NVHS clock on B/E20 (P) and B/F20 (N). Clock circuitry is not included.

J2A

G0_TX3_L5_N	A1	NVHS3_TX_L5_N
G0_TX3_L5_P	A2	NVHS3_TX_L5_P
	A3	GND
G0_TX3_L2_N	A4	NVHS3_TX_L2_N
G0_TX3_L2_P	A5	NVHS3_TX_L2_P
	A6	GND
G0_TX2_L5_N	A7	NVHS2_TX_L5_N
G0_TX2_L5_P	A8	NVHS2_TX_L5_P
	A9	GND
G0_TX2_L2_N	A10	NVHS2_TX_L2_N
G0_TX2_L2_P	A11	NVHS2_TX_L2_P
	A12	GND
	A13	GND
	A14	GND
G0_TX5_L4_N	A15	NVHS5_TX_L4_N
G0_TX5_L4_P	A16	NVHS5_TX_L4_P
	A17	GND
G0_TX5_L1_N	A18	NVHS5_TX_L1_N
G0_TX5_L1_P	A19	NVHS5_TX_L1_P
	A20	GND
G0_TX4_L5_N	A21	NVHS4_TX_L5_N
G0_TX4_L5_P	A22	NVHS4_TX_L5_P
	A23	GND
G0_TX4_L2_N	A24	NVHS4_TX_L2_N
G0_TX4_L2_P	A25	NVHS4_TX_L2_P
	A26	GND
	A27	GND
	A28	GND
G0_TX1_L4_N	A29	NVHS1_TX_L4_N
G0_TX1_L4_P	A30	NVHS1_TX_L4_P
	A31	GND
G0_TX1_L1_N	A32	NVHS1_TX_L1_N
G0_TX1_L1_P	A33	NVHS1_TX_L1_P
	A34	GND
G0_TX0_L5_N	A35	NVHS0_TX_L5_N
G0_TX0_L5_P	A36	NVHS0_TX_L5_P
	A37	GND
G0_TX0_L2_N	A38	NVHS0_TX_L2_N
G0_TX0_L2_P	A39	NVHS0_TX_L2_P
	A40	GND

GPU0 / B / Column A

J2B

	B1	GND
G0_TX3_L4_N	B2	NVHS3_TX_L4_N
G0_TX3_L4_P	B3	NVHS3_TX_L4_P
	B4	GND
G0_TX3_L1_N	B5	NVHS3_TX_L1_N
G0_TX3_L1_P	B6	NVHS3_TX_L1_P
	B7	GND
G0_TX2_L4_N	B8	NVHS2_TX_L4_N
G0_TX2_L4_P	B9	NVHS2_TX_L4_P
	B10	GND
G0_TX2_L1_N	B11	NVHS2_TX_L1_N
G0_TX2_L1_P	B12	NVHS2_TX_L1_P
	B13	GND
G0_TX5_L5_N	B14	NVHS5_TX_L5_N
G0_TX5_L5_P	B15	NVHS5_TX_L5_P
	B16	GND
G0_TX5_L2_N	B17	NVHS5_TX_L2_N
G0_TX5_L2_P	B18	NVHS5_TX_L2_P
	B19	GND
	B20	GND
	B21	GND
G0_TX4_L4_N	B22	NVHS4_TX_L4_N
G0_TX4_L4_P	B23	NVHS4_TX_L4_P
	B24	GND
G0_TX4_L1_N	B25	NVHS4_TX_L1_N
G0_TX4_L1_P	B26	NVHS4_TX_L1_P
	B27	GND
G0_TX1_L5_N	B28	NVHS1_TX_L5_N
G0_TX1_L5_P	B29	NVHS1_TX_L5_P
	B30	GND
G0_TX1_L2_N	B31	NVHS1_TX_L2_N
G0_TX1_L2_P	B32	NVHS1_TX_L2_P
	B33	GND
	B34	GND
	B35	GND
G0_TX0_L4_N	B36	NVHS0_TX_L4_N
G0_TX0_L4_P	B37	NVHS0_TX_L4_P
	B38	GND
G0_TX0_L1_N	B39	NVHS0_TX_L1_N
G0_TX0_L1_P	B40	NVHS0_TX_L1_P

GPU0 / B / Column B

J2C

G0_TX3_L6_N	C1	NVHS3_TX_L6_N
G0_TX3_L6_P	C2	NVHS3_TX_L6_P
	C3	GND
G0_TX3_L3_N	C4	NVHS3_TX_L3_N
G0_TX3_L3_P	C5	NVHS3_TX_L3_P
	C6	GND
G0_TX2_L6_N	C7	NVHS2_TX_L6_N
G0_TX2_L6_P	C8	NVHS2_TX_L6_P
	C9	GND
G0_TX2_L3_N	C10	NVHS2_TX_L3_N
G0_TX2_L3_P	C11	NVHS2_TX_L3_P
	C12	GND
	C13	GND
	C14	GND
G0_TX5_L6_N	C15	NVHS5_TX_L6_N
G0_TX5_L6_P	C16	NVHS5_TX_L6_P
	C17	GND
G0_TX5_L0_N	C18	NVHS5_TX_L0_N
G0_TX5_L0_P	C19	NVHS5_TX_L0_P
	C20	GND
G0_TX4_L6_N	C21	NVHS4_TX_L6_N
G0_TX4_L6_P	C22	NVHS4_TX_L6_P
	C23	GND
G0_TX4_L3_N	C24	NVHS4_TX_L3_N
G0_TX4_L3_P	C25	NVHS4_TX_L3_P
	C26	GND
	C27	GND
	C28	GND
G0_TX1_L6_N	C29	NVHS1_TX_L6_N
G0_TX1_L6_P	C30	NVHS1_TX_L6_P
	C31	GND
G0_TX1_L0_N	C32	NVHS1_TX_L0_N
G0_TX1_L0_P	C33	NVHS1_TX_L0_P
	C34	GND
G0_TX0_L6_N	C35	NVHS0_TX_L6_N
G0_TX0_L6_P	C36	NVHS0_TX_L6_P
	C37	GND
G0_TX0_L3_N	C38	NVHS0_TX_L3_N
G0_TX0_L3_P	C39	NVHS0_TX_L3_P
	C40	GND

GPU0 / B / Column C

J2D

	D1	GND
G0_TX3_L7_N	D2	NVHS3_TX_L7_N
G0_TX3_L7_P	D3	NVHS3_TX_L7_P
	D4	GND
G0_TX3_L0_N	D5	NVHS3_TX_L0_N
G0_TX3_L0_P	D6	NVHS3_TX_L0_P
	D7	GND
G0_TX2_L7_N	D8	NVHS2_TX_L7_N
G0_TX2_L7_P	D9	NVHS2_TX_L7_P
	D10	GND
G0_TX2_L0_N	D11	NVHS2_TX_L0_N
G0_TX2_L0_P	D12	NVHS2_TX_L0_P
	D13	GND
G0_TX5_L7_N	D14	NVHS5_TX_L7_N
G0_TX5_L7_P	D15	NVHS5_TX_L7_P
	D16	GND
G0_TX5_L3_N	D17	NVHS5_TX_L3_N
G0_TX5_L3_P	D18	NVHS5_TX_L3_P
	D19	GND
	D20	GND
	D21	GND
G0_TX4_L7_N	D22	NVHS4_TX_L7_N
G0_TX4_L7_P	D23	NVHS4_TX_L7_P
	D24	GND
G0_TX4_L0_N	D25	NVHS4_TX_L0_N
G0_TX4_L0_P	D26	NVHS4_TX_L0_P
	D27	GND
G0_TX1_L7_N	D28	NVHS1_TX_L7_N
G0_TX1_L7_P	D29	NVHS1_TX_L7_P
	D30	GND
G0_TX1_L3_N	D31	NVHS1_TX_L3_N
G0_TX1_L3_P	D32	NVHS1_TX_L3_P
	D33	GND
	D34	GND
	D35	GND
G0_TX0_L7_N	D36	NVHS0_TX_L7_N
G0_TX0_L7_P	D37	NVHS0_TX_L7_P
	D38	GND
G0_TX0_L0_N	D39	NVHS0_TX_L0_N
G0_TX0_L0_P	D40	NVHS0_TX_L0_P

GPU0 / B / Column D

J2E

	E1	GND
	E2	GND
	E3	GND
	E4	GND
	E5	GND
	E6	GND
	E7	GND
	E8	GND
	E9	GND
	E10	GND
	E11	GND
	E12	GND
	E13	GND
	E14	GND
	E15	GND
	E16	GND
	E17	GND
	E18	GND
	E19	GND
	E20	NVHS_REFCLK_P
	E21	GND
	E22	GND
	E23	GND
	E24	GND
	E25	GND
	E26	GND
	E27	GND
	E28	GND
	E29	GND
	E30	GND
	E31	GND
	E32	GND
	E33	GND
	E34	SMB_CLK_(candidate)
	E35	GND
	E36	GND
	E37	GND
	E38	GND
	E39	GND
	E40	GND

GPU0 / B / Column E

J2F

	F1	GND
	F2	GND
	F3	GND
	F4	GND
	F5	GND
	F6	GND
	F7	GND
	F8	GND
	F9	GND
	F10	GND
	F11	GND
	F12	GND
	F13	GND
	F14	GND
	F15	GND
	F16	GND
	F17	GND
	F18	GND
	F19	GND
	F20	NVHS_REFCLK_N
	F21	GND
	F22	GND
	F23	GND
	F24	GND
	F25	GND
	F26	GND
	F27	GND
	F28	GND
	F29	GND
	F30	GND
	F31	GND
	F32	GND
	F33	GND
	F34	SMB_DAT_(candidate)
	F35	GND
	F36	GND
	F37	GND
	F38	GND
	F39	GND
	F40	GND

GPU0 / B / Column F

J2G

	G1	GND
G3_TX0_L4_N	G2	NVHS3_RX_L4_N
G3_TX0_L4_P	G3	NVHS3_RX_L4_P
	G4	GND
G3_TX0_L1_N	G5	NVHS3_RX_L1_N
G3_TX0_L1_P	G6	NVHS3_RX_L1_P
	G7	GND
G3_TX1_L4_N	G8	NVHS2_RX_L4_N
G3_TX1_L4_P	G9	NVHS2_RX_L4_P
	G10	GND
G3_TX1_L1_N	G11	NVHS2_RX_L1_N
G3_TX1_L1_P	G12	NVHS2_RX_L1_P
	G13	GND
G2_TX4_L5_N	G14	NVHS5_RX_L5_N
G2_TX4_L5_P	G15	NVHS5_RX_L5_P
	G16	GND
G2_TX4_L2_N	G17	NVHS5_RX_L2_N
G2_TX4_L2_P	G18	NVHS5_RX_L2_P
	G19	GND
	G20	GND
	G21	GND
G2_TX5_L4_N	G22	NVHS4_RX_L4_N
G2_TX5_L4_P	G23	NVHS4_RX_L4_P
	G24	GND
G2_TX5_L1_N	G25	NVHS4_RX_L1_N
G2_TX5_L1_P	G26	NVHS4_RX_L1_P
	G27	GND
G1_TX2_L5_N	G28	NVHS1_RX_L5_N
G1_TX2_L5_P	G29	NVHS1_RX_L5_P
	G30	GND
G1_TX2_L2_N	G31	NVHS1_RX_L2_N
G1_TX2_L2_P	G32	NVHS1_RX_L2_P
	G33	GND
	G34	GND
	G35	GND
G1_TX3_L4_N	G36	NVHS0_RX_L4_N
G1_TX3_L4_P	G37	NVHS0_RX_L4_P
	G38	GND
G1_TX3_L1_N	G39	NVHS0_RX_L1_N
G1_TX3_L1_P	G40	NVHS0_RX_L1_P

GPU0 / B / Column G

J2H

G3_TX0_L5_N	H1	NVHS3_RX_L5_N
G3_TX0_L5_P	H2	NVHS3_RX_L5_P
	H3	GND
G3_TX0_L2_N	H4	NVHS3_RX_L2_N
G3_TX0_L2_P	H5	NVHS3_RX_L2_P
	H6	GND
G3_TX1_L5_N	H7	NVHS2_RX_L5_N
G3_TX1_L5_P	H8	NVHS2_RX_L5_P
	H9	GND
G3_TX1_L2_N	H10	NVHS2_RX_L2_N
G3_TX1_L2_P	H11	NVHS2_RX_L2_P
	H12	GND
	H13	PRSNT1_A_(candidate)
	H14	GND
G2_TX4_L4_N	H15	NVHS5_RX_L4_N
G2_TX4_L4_P	H16	NVHS5_RX_L4_P
	H17	GND
G2_TX4_L1_N	H18	NVHS5_RX_L1_N
G2_TX4_L1_P	H19	NVHS5_RX_L1_P
	H20	GND
G2_TX5_L5_N	H21	NVHS4_RX_L5_N
G2_TX5_L5_P	H22	NVHS4_RX_L5_P
	H23	GND
G2_TX5_L2_N	H24	NVHS4_RX_L2_N
G2_TX5_L2_P	H25	NVHS4_RX_L2_P
	H26	GND
	H27	GND
	H28	GND
G1_TX2_L4_N	H29	NVHS1_RX_L4_N
G1_TX2_L4_P	H30	NVHS1_RX_L4_P
	H31	GND
G1_TX2_L1_N	H32	NVHS1_RX_L1_N
G1_TX2_L1_P	H33	NVHS1_RX_L1_P
	H34	GND
G1_TX3_L5_N	H35	NVHS0_RX_L5_N
G1_TX3_L5_P	H36	NVHS0_RX_L5_P
	H37	GND
G1_TX3_L2_N	H38	NVHS0_RX_L2_N
G1_TX3_L2_P	H39	NVHS0_RX_L2_P
	H40	GND

GPU0 / B / Column H

J2J

G3_TX0_L7_N	K1	NVHS3_RX_L7_N
G3_TX0_L7_P	K2	NVHS3_RX_L7_P
	K3	GND
G3_TX0_L3_N	K4	NVHS3_RX_L3_N
G3_TX0_L3_P	K5	NVHS3_RX_L3_P
	K6	GND
G3_TX1_L7_N	K7	NVHS2_RX_L7_N
G3_TX1_L7_P	K8	NVHS2_RX_L7_P
	K9	GND
G3_TX1_L3_N	K10	NVHS2_RX_L3_N
G3_TX1_L3_P	K11	NVHS2_RX_L3_P
	K12	GND
	K13	GND
	K14	GND
G2_TX4_L7_N	K15	NVHS5_RX_L7_N
G2_TX4_L7_P	K16	NVHS5_RX_L7_P
	K17	GND
G2_TX4_L0_N	K18	NVHS5_RX_L0_N
G2_TX4_L0_P	K19	NVHS5_RX_L0_P
	K20	GND
G2_TX5_L7_N	K21	NVHS4_RX_L7_N
G2_TX5_L7_P	K22	NVHS4_RX_L7_P
	K23	GND
G2_TX5_L3_N	K24	NVHS4_RX_L3_N
G2_TX5_L3_P	K25	NVHS4_RX_L3_P
	K26	GND
	K27	GND
	K28	GND
G1_TX2_L7_N	K29	NVHS1_RX_L7_N
G1_TX2_L7_P	K30	NVHS1_RX_L7_P
	K31	GND
G1_TX2_L0_N	K32	NVHS1_RX_L0_N
G1_TX2_L0_P	K33	NVHS1_RX_L0_P
	K34	GND
G1_TX3_L7_N	K35	NVHS0_RX_L7_N
G1_TX3_L7_P	K36	NVHS0_RX_L7_P
	K37	GND
G1_TX3_L3_N	K38	NVHS0_RX_L3_N
G1_TX3_L3_P	K39	NVHS0_RX_L3_P
	K40	GND

GPU0 / B / Column K

NET LABEL KEY / Example: G0_TX0_L0_P

G0
Transmitting GPU0

TX0
NVHS0 transmit

L0
Lane 0

P / N
Positive / negative

Both ends use the transmitting contact name: full signal names remain beside every pin.
Only NVLink data is wired. An unconnected contact is not an official NVIDIA NC assignment.

Logical NVLink reference: no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU0 / J1 + J2/GPU0 Connector B / J2/
File: GPU0_Connector_B.kicad_sch

Title: GPU0 — Connector B — 400 contacts

Size: A2	Date: 2026-10-30
KiCad E.D.A. 10.0.7	**Rev: Reference 1**
	Id: 4/13

GPU1: 800 labelled contacts / 192 NVLink data contacts connected
Open Connector A for PCIe, clock and power/control contacts; open B for the NVLink net labels.

GPU1 Connector A / J3



File: GPU1_Connector_A.kicad_sch

GPU1 Connector B / J4



File: GPU1_Connector_B.kicad_sch

Supporting clock, supply and control pins remain labelled and unconnected.
Contact symbols are passive and have no assigned PCB footprint.

CLOCK CIRCUITRY REQUIRED FOR OPERATION
NVLink: 156.25 MHz differential / B:E20=P, B:F20=N
PCIe: 100 MHz differential / A:E7=P, A:F7=N

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU1 / J3 + J4/
File: GPU1.kicad_sch

Title: GPU1 – V100 SXM2 connectors

Size: A4	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 5/13

All 400 contacts retain their signal names and pad numbers. Clocks, supplies, PCIe, reset and control circuits are left for your design.

J3A

A1	GND
A2	PERp0
A3	PERn0
A4	GND
A5	PERp7
A6	PERn7
A7	GND
A8	PERp8
A9	PERn8
A10	GND
A11	PERp12
A12	PERn12
A13	GND
A14	GND
A15	UNRESOLVED/NC
A16	UNRESOLVED/NC
A17	GND
A18	PWR_GOOD
A19	GND
A20	+5V_AUX
A21	GND
A22	12V
A23	12V
A24	GND
A25	12V
A26	12V
A27	GND
A28	12V
A29	12V
A30	GND
A31	12V
A32	12V
A33	GND
A34	12V
A35	12V
A36	GND
A37	12V
A38	12V
A39	GND
A40	12V

GPU1 / A / Column A

J3B

B1	PERp1
B2	PERn1
B3	GND
B4	PERp4
B5	PERn4
B6	GND
B7	GND
B8	GND
B9	PERp9
B10	PERn9
B11	GND
B12	PERp15
B13	PERn15
B14	GND
B15	GND
B16	UNRESOLVED/NC
B17	UNRESOLVED/NC
B18	GND
B19	GND
B20	+5V_AUX
B21	GND
B22	12V
B23	12V
B24	GND
B25	12V
B26	12V
B27	GND
B28	12V
B29	12V
B30	GND
B31	12V
B32	12V
B33	GND
B34	12V
B35	12V
B36	GND
B37	12V
B38	12V
B39	GND
B40	12V

GPU1 / A / Column B

J3C

C1	GND
C2	PERp3
C3	PERn3
C4	GND
C5	PERp6
C6	PERn6
C7	GND
C8	PERp10
C9	PERn10
C10	GND
C11	PERp13
C12	PERn13
C13	GND
C14	GND
C15	UNRESOLVED/NC
C16	UNRESOLVED/NC
C17	GND
C18	I2C_ADDR0
C19	GND
C20	+5V_AUX
C21	GND
C22	12V
C23	12V
C24	GND
C25	12V
C26	12V
C27	GND
C28	12V
C29	12V
C30	GND
C31	12V
C32	12V
C33	GND
C34	12V
C35	12V
C36	GND
C37	12V
C38	12V
C39	GND
C40	12V

GPU1 / A / Column C

J3D

D1	PERp2
D2	PERn2
D3	GND
D4	PERp5
D5	PERn5
D6	GND
D7	GND
D8	GND
D9	PERp11
D10	PERn11
D11	GND
D12	PERp14
D13	PERn14
D14	GND
D15	GND
D16	UNRESOLVED/NC
D17	UNRESOLVED/NC
D18	GND
D19	GND
D20	+5V_AUX
D21	GND
D22	12V
D23	12V
D24	GND
D25	12V
D26	12V
D27	GND
D28	12V
D29	12V
D30	GND
D31	12V
D32	12V
D33	GND
D34	12V
D35	12V
D36	GND
D37	12V
D38	12V
D39	GND
D40	12V

GPU1 / A / Column D

J3E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	PCIE_REFCLK_P
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	PERST#
E19	GND
E20	+5V_AUX
E21	GND
E22	12V
E23	12V
E24	GND
E25	12V
E26	12V
E27	GND
E28	12V
E29	12V
E30	GND
E31	12V
E32	12V
E33	GND
E34	12V
E35	12V
E36	GND
E37	12V
E38	12V
E39	GND
E40	12V

GPU1 / A / Column E

J3F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	PCIE_REFCLK_N
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	PWR_BRAKE#
F19	GND
F20	JTAG_TDI
F21	GND
F22	12V
F23	12V
F24	GND
F25	12V
F26	12V
F27	GND
F28	12V
F29	12V
F30	GND
F31	12V
F32	12V
F33	GND
F34	12V
F35	12V
F36	GND
F37	12V
F38	12V
F39	GND
F40	12V

GPU1 / A / Column F

J3G

G1	PETp0
G2	PETn0
G3	GND
G4	PETp4
G5	PETn4
G6	GND
G7	GND
G8	GND
G9	PETp8
G10	PETn8
G11	GND
G12	PETp15
G13	PETn15
G14	GND
G15	GND
G16	UNRESOLVED/NC
G17	UNRESOLVED/NC
G18	GND
G19	GND
G20	JTAG_TDO
G21	GND
G22	12V
G23	12V
G24	GND
G25	12V
G26	12V
G27	GND
G28	12V
G29	12V
G30	GND
G31	12V
G32	12V
G33	GND
G34	12V
G35	12V
G36	GND
G37	12V
G38	12V
G39	GND
G40	12V

GPU1 / A / Column G

J3H

H1	GND
H2	PETp1
H3	PETn1
H4	GND
H5	PETp7
H6	PETn7
H7	GND
H8	PETp9
H9	PETn9
H10	GND
H11	PETp12
H12	PETn12
H13	GND
H14	GND
H15	UNRESOLVED/NC
H16	UNRESOLVED/NC
H17	GND
H18	I2C_ADDR1
H19	JTAG_SEL
H20	JTAG_TCLK
H21	GND
H22	12V
H23	12V
H24	GND
H25	12V
H26	12V
H27	GND
H28	12V
H29	12V
H30	GND
H31	12V
H32	12V
H33	GND
H34	12V
H35	12V
H36	GND
H37	12V
H38	12V
H39	GND
H40	12V

GPU1 / A / Column H

J3I

J1	PETp2
J2	PETn2
J3	GND
J4	PETp5
J5	PETn5
J6	GND
J7	GND
J8	GND
J9	PETp11
J10	PETn11
J11	GND
J12	PETp14
J13	PETn14
J14	GND
J15	GND
J16	UNRESOLVED/NC
J17	UNRESOLVED/NC
J18	GND
J19	GND
J20	JTAG_TMS
J21	GND
J22	12V
J23	12V
J24	GND
J25	12V
J26	12V
J27	GND
J28	12V
J29	12V
J30	GND
J31	12V
J32	12V
J33	GND
J34	12V
J35	12V
J36	GND
J37	12V
J38	12V
J39	GND
J40	12V

GPU1 / A / Column J

J3J

K1	GND
K2	PETp3
K3	PETn3
K4	GND
K5	PETp6
K6	PETn6
K7	GND
K8	PETp10
K9	PETn10
K10	GND
K11	PETp13
K12	PETn13
K13	GND
K14	GND
K15	UNRESOLVED/NC
K16	UNRESOLVED/NC
K17	GND
K18	PWR_EN
K19	THERM_OVERT#
K20	JTAG_TRST#
K21	GND
K22	12V
K23	12V
K24	GND
K25	12V
K26	12V
K27	GND
K28	12V
K29	12V
K30	GND
K31	12V
K32	12V
K33	GND
K34	12V
K35	12V
K36	GND
K37	12V
K38	12V
K39	GND
K40	12V

GPU1 / A / Column K

Separate PCIe reference clock: 100 MHz differential on A/E7 (P) and A/F7 (N). Clock circuitry is not included.

A/B7 and A/A39: disputed GND labels (reference presence loop, PRSNT2_A/B).
Ground both only if your carrier omits presence detection and its controller does not depend on it.
For presence detection, implement and validate the loop. Grounding its readout can force a false present state.
Source: Quanta Big Basin sheets 15–16. See docs/presence-detection.md.

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

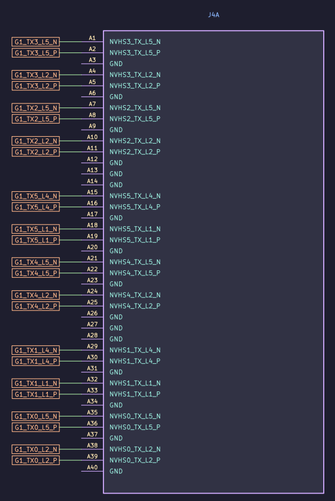
Sheet: /GPU1 / J3 + J4/GPU1 Connector A / J3/
File: GPU1_Connector_A.kicad_sch

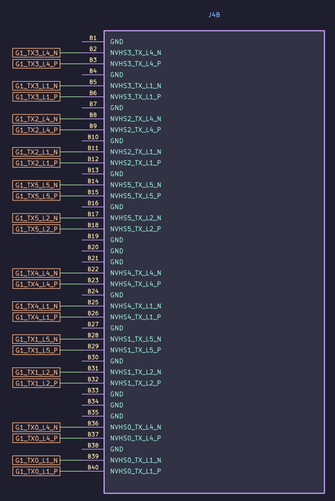
Title: GPU1 — Connector A — 400 contacts

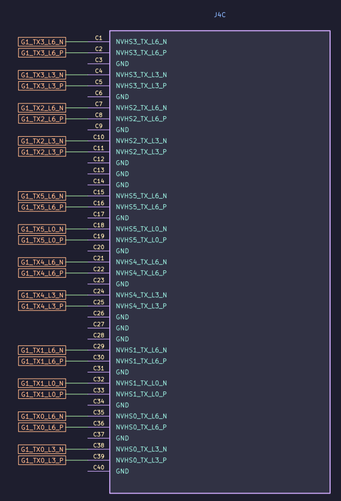
Size: A3	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 6/13

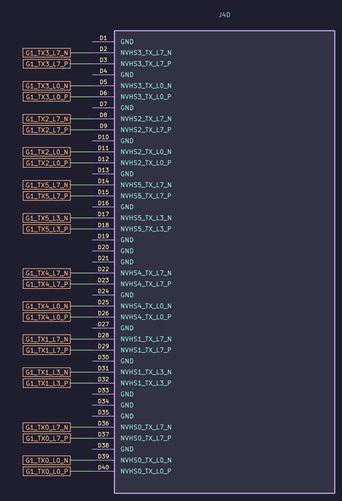
NVLink data pins connect through matching global labels. Clock, ground, power and control contacts keep their names and remain unconnected.

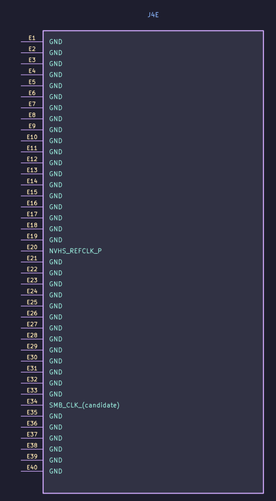
DATA WIRING ONLY — EXTERNAL CLOCK REQUIRED
Each V100: 156.25 MHz differential NVHS clock on B/E20 (P) and B/F20 (N). Clock circuitry is not included.

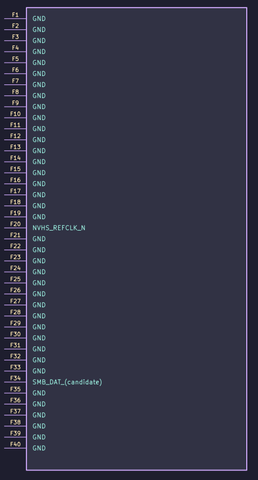


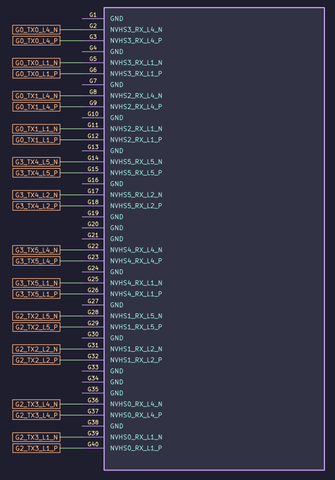


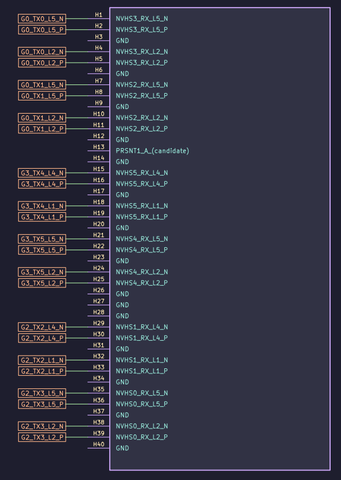


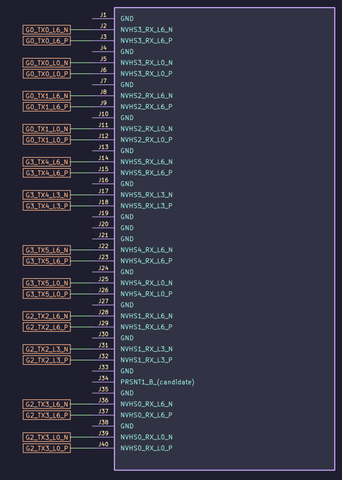


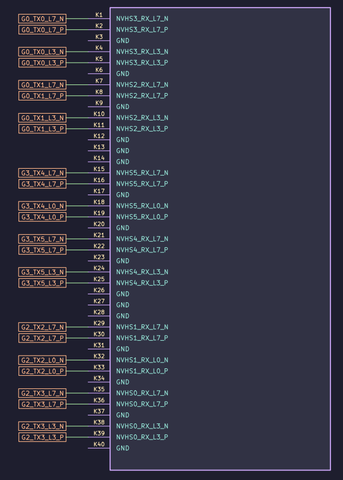












NET LABEL KEY / Example: G0_TX0_L0_P

G0
Transmitting GPU0

TX0
NVHS0 transmit

L0
Lane 0

P / N
Positive / negative

Both ends use the transmitting contact name: full signal names remain beside every pin.
Only NVLink data is wired. An unconnected contact is not an official NVIDIA NC assignment.

Logical NVLink reference: no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU1 / J3 + J4/GPU1 Connector B / J4/
File: GPU1_Connector_B.kicad_sch

Title: GPU1 — Connector B — 400 contacts

Size: A2	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 7/13

GPU2: 800 labelled contacts / 192 NVLink data contacts connected
Open Connector A for PCIe, clock and power/control contacts; open B for the NVLink net labels.

GPU2 Connector A / J5



File: GPU2_Connector_A.kicad_sch

GPU2 Connector B / J6



File: GPU2_Connector_B.kicad_sch

Supporting clock, supply and control pins remain labelled and unconnected.
Contact symbols are passive and have no assigned PCB footprint.

CLOCK CIRCUITRY REQUIRED FOR OPERATION
NVLink: 156.25 MHz differential / B:E20=P, B:F20=N
PCIe: 100 MHz differential / A:E7=P, A:F7=N

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU2 / J5 + J6/
File: GPU2.kicad_sch

Title: GPU2 — V100 SXM2 connectors

Size: A4	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 8/13

All 400 contacts retain their signal names and pad numbers. Clocks, supplies, PCIe, reset and control circuits are left for your design.

J5A

A1	GND
A2	PERp0
A3	PERn0
A4	GND
A5	PERp7
A6	PERn7
A7	GND
A8	PERp8
A9	PERn8
A10	GND
A11	PERp12
A12	PERn12
A13	GND
A14	GND
A15	UNRESOLVED/NC
A16	UNRESOLVED/NC
A17	GND
A18	PWR_GOOD
A19	GND
A20	+5V_AUX
A21	GND
A22	12V
A23	12V
A24	GND
A25	12V
A26	12V
A27	GND
A28	12V
A29	12V
A30	GND
A31	12V
A32	12V
A33	GND
A34	12V
A35	12V
A36	GND
A37	12V
A38	12V
A39	GND
A40	12V

GPU2 / A / Column A

J5B

B1	PERp1
B2	PERn1
B3	GND
B4	PERp4
B5	PERn4
B6	GND
B7	GND
B8	GND
B9	PERp9
B10	PERn9
B11	GND
B12	PERp15
B13	PERn15
B14	GND
B15	GND
B16	UNRESOLVED/NC
B17	UNRESOLVED/NC
B18	GND
B19	GND
B20	+5V_AUX
B21	GND
B22	12V
B23	12V
B24	GND
B25	12V
B26	12V
B27	GND
B28	12V
B29	12V
B30	GND
B31	12V
B32	12V
B33	GND
B34	12V
B35	12V
B36	GND
B37	12V
B38	12V
B39	GND
B40	12V

GPU2 / A / Column B

J5C

C1	GND
C2	PERp3
C3	PERn3
C4	GND
C5	PERp6
C6	PERn6
C7	GND
C8	PERp10
C9	PERn10
C10	GND
C11	PERp13
C12	PERn13
C13	GND
C14	GND
C15	UNRESOLVED/NC
C16	UNRESOLVED/NC
C17	GND
C18	I2C_ADDR0
C19	GND
C20	+5V_AUX
C21	GND
C22	12V
C23	12V
C24	GND
C25	12V
C26	12V
C27	GND
C28	12V
C29	12V
C30	GND
C31	12V
C32	12V
C33	GND
C34	12V
C35	12V
C36	GND
C37	12V
C38	12V
C39	GND
C40	12V

GPU2 / A / Column C

J5D

D1	PERp2
D2	PERn2
D3	GND
D4	PERp5
D5	PERn5
D6	GND
D7	GND
D8	GND
D9	PERp11
D10	PERn11
D11	GND
D12	PERp14
D13	PERn14
D14	GND
D15	GND
D16	UNRESOLVED/NC
D17	UNRESOLVED/NC
D18	GND
D19	GND
D20	+5V_AUX
D21	GND
D22	12V
D23	12V
D24	GND
D25	12V
D26	12V
D27	GND
D28	12V
D29	12V
D30	GND
D31	12V
D32	12V
D33	GND
D34	12V
D35	12V
D36	GND
D37	12V
D38	12V
D39	GND
D40	12V

GPU2 / A / Column D

J5E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	PCIE_REFCLK_P
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	PERST#
E19	GND
E20	+5V_AUX
E21	GND
E22	12V
E23	12V
E24	GND
E25	12V
E26	12V
E27	GND
E28	12V
E29	12V
E30	GND
E31	12V
E32	12V
E33	GND
E34	12V
E35	12V
E36	GND
E37	12V
E38	12V
E39	GND
E40	12V

GPU2 / A / Column E

J5F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	PCIE_REFCLK_N
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	PWR_BRAKE#
F19	GND
F20	JTAG_TDI
F21	GND
F22	12V
F23	12V
F24	GND
F25	12V
F26	12V
F27	GND
F28	12V
F29	12V
F30	GND
F31	12V
F32	12V
F33	GND
F34	12V
F35	12V
F36	GND
F37	12V
F38	12V
F39	GND
F40	12V

GPU2 / A / Column F

J5G

G1	PETp0
G2	PETn0
G3	GND
G4	PETp4
G5	PETn4
G6	GND
G7	GND
G8	GND
G9	PETp8
G10	PETn8
G11	GND
G12	PETp15
G13	PETn15
G14	GND
G15	GND
G16	UNRESOLVED/NC
G17	UNRESOLVED/NC
G18	GND
G19	GND
G20	JTAG_TDO
G21	GND
G22	12V
G23	12V
G24	GND
G25	12V
G26	12V
G27	GND
G28	12V
G29	12V
G30	GND
G31	12V
G32	12V
G33	GND
G34	12V
G35	12V
G36	GND
G37	12V
G38	12V
G39	GND
G40	12V

GPU2 / A / Column G

J5H

H1	GND
H2	PETp1
H3	PETn1
H4	GND
H5	PETp7
H6	PETn7
H7	GND
H8	PETp9
H9	PETn9
H10	GND
H11	PETp12
H12	PETn12
H13	GND
H14	GND
H15	UNRESOLVED/NC
H16	UNRESOLVED/NC
H17	GND
H18	I2C_ADDR1
H19	JTAG_SEL
H20	JTAG_TCLK
H21	GND
H22	12V
H23	12V
H24	GND
H25	12V
H26	12V
H27	GND
H28	12V
H29	12V
H30	GND
H31	12V
H32	12V
H33	GND
H34	12V
H35	12V
H36	GND
H37	12V
H38	12V
H39	GND
H40	12V

GPU2 / A / Column H

J5I

J1	PETp2
J2	PETn2
J3	GND
J4	PETp5
J5	PETn5
J6	GND
J7	GND
J8	GND
J9	PETp11
J10	PETn11
J11	GND
J12	PETp14
J13	PETn14
J14	GND
J15	GND
J16	UNRESOLVED/NC
J17	UNRESOLVED/NC
J18	GND
J19	GND
J20	JTAG_TMS
J21	GND
J22	12V
J23	12V
J24	GND
J25	12V
J26	12V
J27	GND
J28	12V
J29	12V
J30	GND
J31	12V
J32	12V
J33	GND
J34	12V
J35	12V
J36	GND
J37	12V
J38	12V
J39	GND
J40	12V

GPU2 / A / Column J

J5J

K1	GND
K2	PETp3
K3	PETn3
K4	GND
K5	PETp6
K6	PETn6
K7	GND
K8	PETp10
K9	PETn10
K10	GND
K11	PETp13
K12	PETn13
K13	GND
K14	GND
K15	UNRESOLVED/NC
K16	UNRESOLVED/NC
K17	GND
K18	PWR_EN
K19	THERM_OVERT#
K20	JTAG_TRST#
K21	GND
K22	12V
K23	12V
K24	GND
K25	12V
K26	12V
K27	GND
K28	12V
K29	12V
K30	GND
K31	12V
K32	12V
K33	GND
K34	12V
K35	12V
K36	GND
K37	12V
K38	12V
K39	GND
K40	12V

GPU2 / A / Column K

Separate PCIe reference clock: 100 MHz differential on A/E7 (P) and A/F7 (N). Clock circuitry is not included.

A/B7 and A/A39: disputed GND labels (reference presence loop, PRSNT2_A/B).
Ground both only if your carrier omits presence detection and its controller does not depend on it.
For presence detection, implement and validate the loop. Grounding its readout can force a false present state.
Source: Quanta Big Basin sheets 15–16. See docs/presence-detection.md.

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

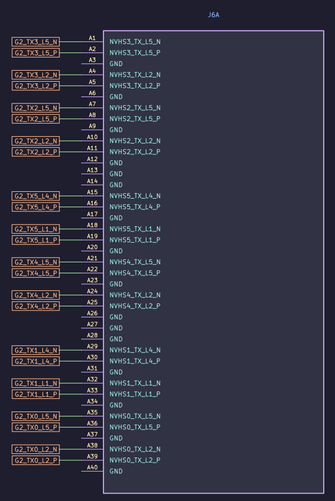
Sheet: /GPU2 / J5 + J6/GPU2 Connector A / J5/
File: GPU2_Connector_A.kicad_sch

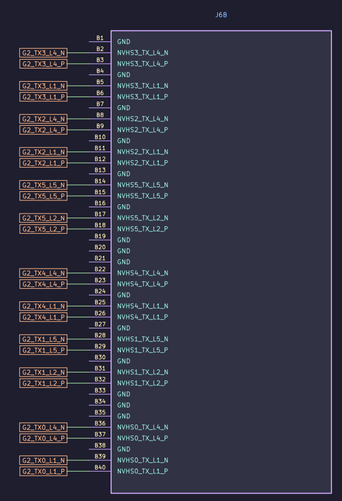
Title: GPU2 — Connector A — 400 contacts

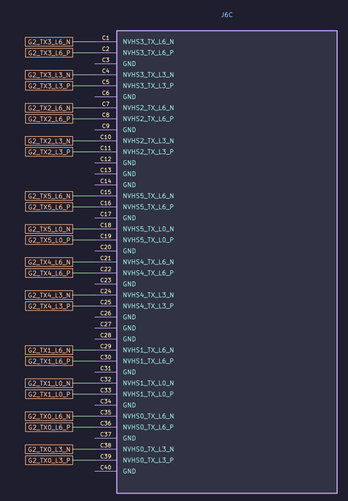
Size: A3	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 9/13

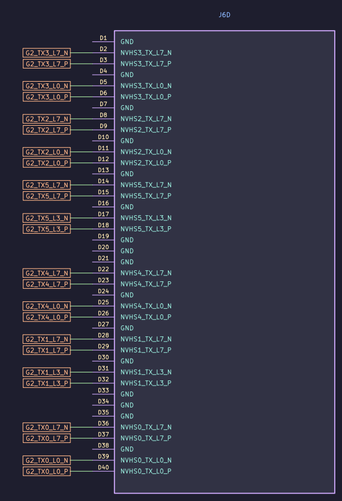
NVLink data pins connect through matching global labels. Clock, ground, power and control contacts keep their names and remain unconnected.

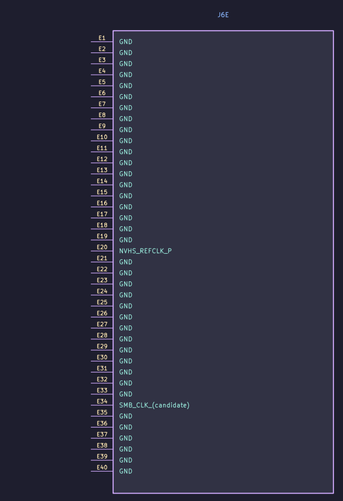
DATA WIRING ONLY — EXTERNAL CLOCK REQUIRED
Each V100: 156.25 MHz differential NVHS clock on B/E20 (P) and B/F20 (N). Clock circuitry is not included.

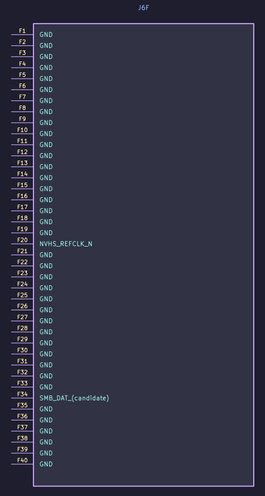


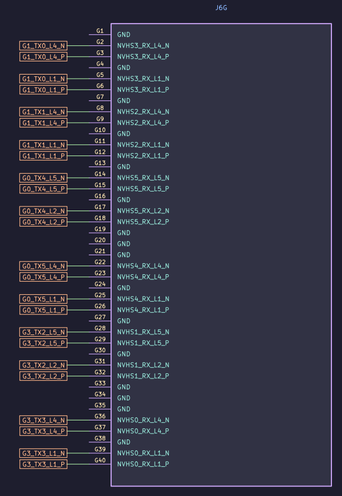


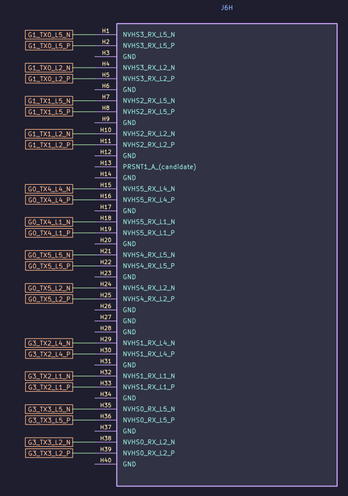


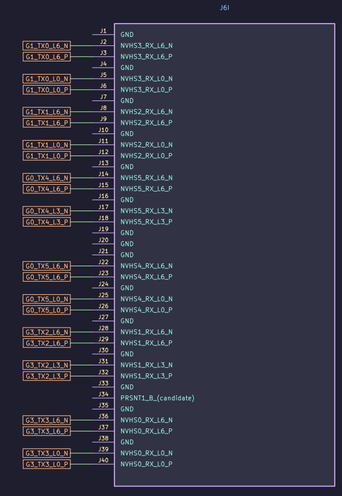


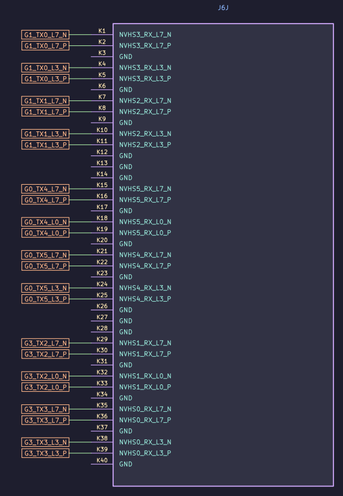












NET LABEL KEY / Example: G0_TX0_L0_P

G0
Transmitting GPU0

TX0
NVHS0 transmit

L0
Lane 0

P / N
Positive / negative

Both ends use the transmitting contact name: full signal names remain beside every pin.
Only NVLink data is wired. An unconnected contact is not an official NVIDIA NC assignment.

Logical NVLink reference: no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU2 / J5 + J6/GPU2 Connector B / J6/
File: GPU2_Connector_B.kicad_sch

Title: GPU2 — Connector B — 400 contacts

Size: A2	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 10/13

GPU3: 800 labelled contacts / 192 NVLink data contacts connected
Open Connector A for PCIe, clock and power/control contacts; open B for the NVLink net labels.

GPU3 Connector A / J7



File: GPU3_Connector_A.kicad_sch

GPU3 Connector B / J8



File: GPU3_Connector_B.kicad_sch

Supporting clock, supply and control pins remain labelled and unconnected.
Contact symbols are passive and have no assigned PCB footprint.

CLOCK CIRCUITRY REQUIRED FOR OPERATION
NVLink: 156.25 MHz differential / B:E20=P, B:F20=N
PCIe: 100 MHz differential / A:E7=P, A:F7=N

Logical NVLink reference; no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU3 / J7 + J8/
File: GPU3.kicad_sch

Title: GPU3 — V100 SXM2 connectors

Size: A4	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 11/13

All 400 contacts retain their signal names and pad numbers. Clocks, supplies, PCIe, reset and control circuits are left for your design.

J7A

A1	GND
A2	PERp0
A3	PERn0
A4	GND
A5	PERp7
A6	PERn7
A7	GND
A8	PERp8
A9	PERn8
A10	GND
A11	PERp12
A12	PERn12
A13	GND
A14	GND
A15	UNRESOLVED/NC
A16	UNRESOLVED/NC
A17	GND
A18	PWR_GOOD
A19	GND
A20	+5V_AUX
A21	GND
A22	12V
A23	12V
A24	GND
A25	12V
A26	12V
A27	GND
A28	12V
A29	12V
A30	GND
A31	12V
A32	12V
A33	GND
A34	12V
A35	12V
A36	GND
A37	12V
A38	12V
A39	GND
A40	12V

GPU3 / A / Column A

J7B

B1	PERp1
B2	PERn1
B3	GND
B4	PERp4
B5	PERn4
B6	GND
B7	GND
B8	GND
B9	PERp9
B10	PERn9
B11	GND
B12	PERp15
B13	PERn15
B14	GND
B15	GND
B16	UNRESOLVED/NC
B17	UNRESOLVED/NC
B18	GND
B19	GND
B20	+5V_AUX
B21	GND
B22	12V
B23	12V
B24	GND
B25	12V
B26	12V
B27	GND
B28	12V
B29	12V
B30	GND
B31	12V
B32	12V
B33	GND
B34	12V
B35	12V
B36	GND
B37	12V
B38	12V
B39	GND
B40	12V

GPU3 / A / Column B

J7C

C1	GND
C2	PERp3
C3	PERn3
C4	GND
C5	PERp6
C6	PERn6
C7	GND
C8	PERp10
C9	PERn10
C10	GND
C11	PERp13
C12	PERn13
C13	GND
C14	GND
C15	UNRESOLVED/NC
C16	UNRESOLVED/NC
C17	GND
C18	I2C_ADDR0
C19	GND
C20	+5V_AUX
C21	GND
C22	12V
C23	12V
C24	GND
C25	12V
C26	12V
C27	GND
C28	12V
C29	12V
C30	GND
C31	12V
C32	12V
C33	GND
C34	12V
C35	12V
C36	GND
C37	12V
C38	12V
C39	GND
C40	12V

GPU3 / A / Column C

J7D

D1	PERp2
D2	PERn2
D3	GND
D4	PERp5
D5	PERn5
D6	GND
D7	GND
D8	GND
D9	PERp11
D10	PERn11
D11	GND
D12	PERp14
D13	PERn14
D14	GND
D15	GND
D16	UNRESOLVED/NC
D17	UNRESOLVED/NC
D18	GND
D19	GND
D20	+5V_AUX
D21	GND
D22	12V
D23	12V
D24	GND
D25	12V
D26	12V
D27	GND
D28	12V
D29	12V
D30	GND
D31	12V
D32	12V
D33	GND
D34	12V
D35	12V
D36	GND
D37	12V
D38	12V
D39	GND
D40	12V

GPU3 / A / Column D

J7E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	PCIE_REFCLK_P
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	PERST#
E19	GND
E20	+5V_AUX
E21	GND
E22	12V
E23	12V
E24	GND
E25	12V
E26	12V
E27	GND
E28	12V
E29	12V
E30	GND
E31	12V
E32	12V
E33	GND
E34	12V
E35	12V
E36	GND
E37	12V
E38	12V
E39	GND
E40	12V

GPU3 / A / Column E

J7F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	PCIE_REFCLK_N
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	PWR_BRAKE#
F19	GND
F20	JTAG_TDI
F21	GND
F22	12V
F23	12V
F24	GND
F25	12V
F26	12V
F27	GND
F28	12V
F29	12V
F30	GND
F31	12V
F32	12V
F33	GND
F34	12V
F35	12V
F36	GND
F37	12V
F38	12V
F39	GND
F40	12V

GPU3 / A / Column F

J7G

G1	PETp0
G2	PETn0
G3	GND
G4	PETp4
G5	PETn4
G6	GND
G7	GND
G8	GND
G9	PETp8
G10	PETn8
G11	GND
G12	PETp15
G13	PETn15
G14	GND
G15	GND
G16	UNRESOLVED/NC
G17	UNRESOLVED/NC
G18	GND
G19	GND
G20	JTAG_TDO
G21	GND
G22	12V
G23	12V
G24	GND
G25	12V
G26	12V
G27	GND
G28	12V
G29	12V
G30	GND
G31	12V
G32	12V
G33	GND
G34	12V
G35	12V
G36	GND
G37	12V
G38	12V
G39	GND
G40	12V

GPU3 / A / Column G

J7H

H1	GND
H2	PETp1
H3	PETn1
H4	GND
H5	PETp7
H6	PETn7
H7	GND
H8	PETp9
H9	PETn9
H10	GND
H11	PETp12
H12	PETn12
H13	GND
H14	GND
H15	UNRESOLVED/NC
H16	UNRESOLVED/NC
H17	GND
H18	I2C_ADDR1
H19	JTAG_SEL
H20	JTAG_TCLK
H21	GND
H22	12V
H23	12V
H24	GND
H25	12V
H26	12V
H27	GND
H28	12V
H29	12V
H30	GND
H31	12V
H32	12V
H33	GND
H34	12V
H35	12V
H36	GND
H37	12V
H38	12V
H39	GND
H40	12V

GPU3 / A / Column H

J7I

J1	PETp2
J2	PETn2
J3	GND
J4	PETp5
J5	PETn5
J6	GND
J7	GND
J8	GND
J9	PETp11
J10	PETn11
J11	GND
J12	PETp14
J13	PETn14
J14	GND
J15	GND
J16	UNRESOLVED/NC
J17	UNRESOLVED/NC
J18	GND
J19	GND
J20	JTAG_TMS
J21	GND
J22	12V
J23	12V
J24	GND
J25	12V
J26	12V
J27	GND
J28	12V
J29	12V
J30	GND
J31	12V
J32	12V
J33	GND
J34	12V
J35	12V
J36	GND
J37	12V
J38	12V
J39	GND
J40	12V

GPU3 / A / Column J

J7J

K1	GND
K2	PETp3
K3	PETn3
K4	GND
K5	PETp6
K6	PETn6
K7	GND
K8	PETp10
K9	PETn10
K10	GND
K11	PETp13
K12	PETn13
K13	GND
K14	GND
K15	UNRESOLVED/NC
K16	UNRESOLVED/NC
K17	GND
K18	PWR_EN
K19	THERM_OVERT#
K20	JTAG_TRST#
K21	GND
K22	12V
K23	12V
K24	GND
K25	12V
K26	12V
K27	GND
K28	12V
K29	12V
K30	GND
K31	12V
K32	12V
K33	GND
K34	12V
K35	12V
K36	GND
K37	12V
K38	12V
K39	GND
K40	12V

GPU3 / A / Column K

Separate PCIe reference clock: 100 MHz differential on A/E7 (P) and A/F7 (N). Clock circuitry is not included.

A/B7 and A/A39: disputed GND labels (reference presence loop, PRSNT2_A/B).
Ground both only if your carrier omits presence detection and its controller does not depend on it.
For presence detection, implement and validate the loop. Grounding its readout can force a false present state.
Source: Quanta Big Basin sheets 15–16. See docs/presence-detection.md.

Logical NVLink reference; no hardware qualification or footprint assignment

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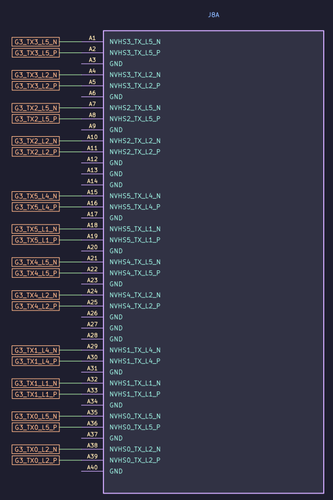
Sheet: /GPU3 / J7 + J8/GPU3 Connector A / J7/
File: GPU3_Connector_A.kicad_sch

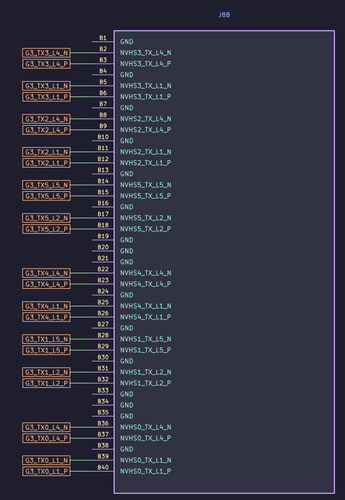
Title: GPU3 — Connector A — 400 contacts

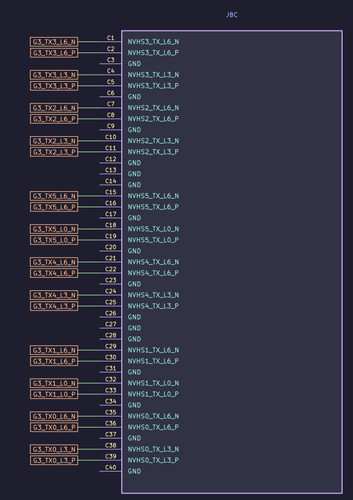
Size: A3	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 12/13

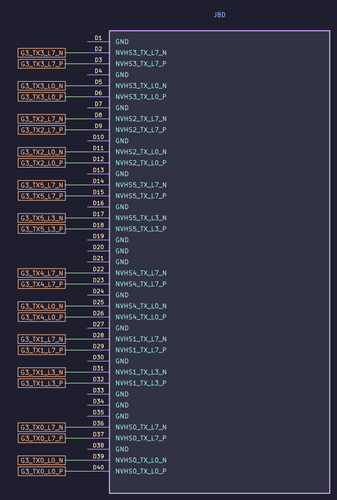
NVLink data pins connect through matching global labels. Clock, ground, power and control contacts keep their names and remain unconnected.

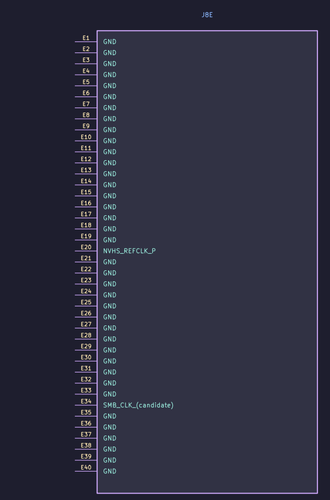
DATA WIRING ONLY — EXTERNAL CLOCK REQUIRED
Each V100: 156.25 MHz differential NVHS clock on B/E20 (P) and B/F20 (N). Clock circuitry is not included.

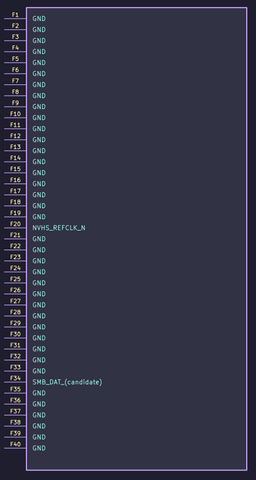


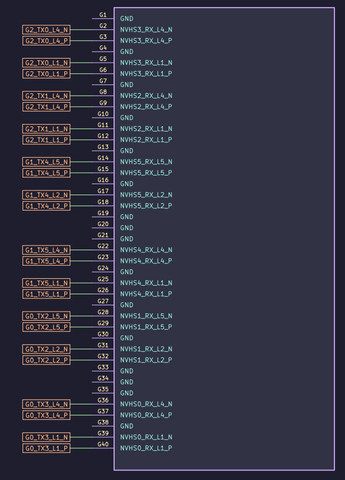


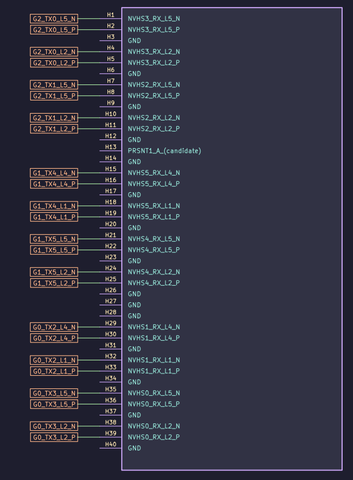


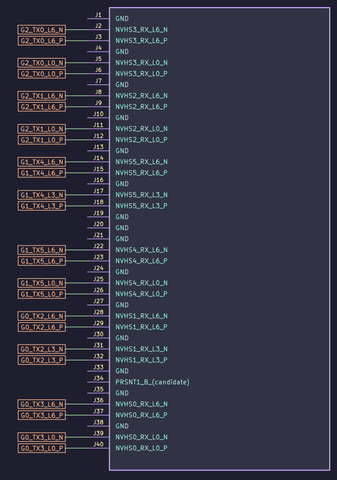


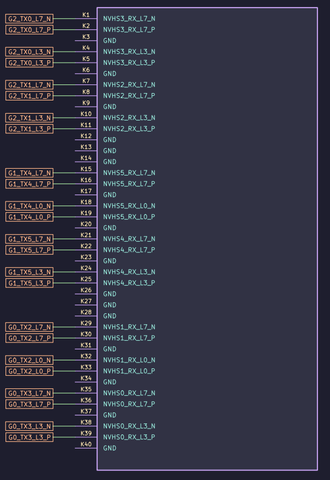












NET LABEL KEY / Example: G0_TX0_L0_P

G0
Transmitting GPU0

TX0
NVHS0 transmit

L0
Lane 0

P / N
Positive / negative

Both ends use the transmitting contact name: full signal names remain beside every pin.
Only NVLink data is wired. An unconnected contact is not an official NVIDIA NC assignment.

Logical NVLink reference: no hardware qualification or footprint assignment

Pigeon Labs

Sheet: /GPU3 / J7 + J8/GPU3 Connector B / J8/
File: GPU3_Connector_B.kicad_sch

Title: GPU3 — Connector B — 400 contacts

Size: A2	Date: 2026-10-30	**Rev: Reference 1**
KiCad E.D.A. 10.0.7		Id: 13/13